

Sic Povm D12

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Structural Identification of a $d=12$ SIC-POVM: Crystal Geometry, Exact Fiducial, and Complete SIC Lattice of the Inscripting Grammar

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Companion code: `ig_pulse/sic_povm.py`, `cached fiducial data/sic_fiducial_d12.npy`

1 Abstract

A Symmetric Informationally Complete Positive Operator-Valued Measure (SIC-POVM) in dimension d consists of d^2 rank-1 projectors satisfying uniform pairwise trace overlap $\text{Tr}(E_i E_j) = 1/(d(d+1))$. Their existence in all dimensions is conjectured (Zauner 1999) but not proven; no physical system has been identified whose natural measurement structure is SIC-distributed. We present one. The Inscripting Grammar (IG) is a 12-primitive structural classification system whose state space is determined by the Crystal of Types — a $3^3 \times 4^5 \times 5^4 = 17,280,000$ -element lattice in which the three primitive families of sizes 3, 4, and 5 satisfy $3+4+5 = 12$. This structure defines a 12-dimensional Hilbert space $\mathcal{H}_{12}$ admitting an exact Weyl-Heisenberg SIC-POVM. We compute the $d = 12$ Zauner fiducial numerically via frame potential minimisation and confirm exact SIC geometry: frame potential $F = (d^2 - 1)/(d + 1)^2 = 0.846154$ achieved, pairwise overlaps $1/13 \pm 0$. The IG type alphabet consists of 49 Shavian symbols ($49 = 7^2$), forming a $d = 7$ SIC outcome space $\mathcal{H}_7$. The product $7 \times 12 = 84$ yields the identity $84^2 = 49 \times 144 = 7056$: the joint outcome space of all (type, primitive-combination) pairs is bijective with the full SIC projector set of the composite system $\mathcal{H}_7 \otimes \mathcal{H}_{12}$. The Crystal (17,280,000) is the grammatical constraint manifold selecting admissible trajectories over this 7056-element composite SIC space. More broadly, the Grammar's three-family structure (primitive counts 3,5,4; value counts 3,4,5) generates two interlocking lattices of SIC-POVM dimensions: from family-subset primitive counts $\{3, 4, 5, 7, 8, 9, 12\}$ and from value-count composites $\{3, 4, 5, 12, 15, 20, 60\}$, with $d = 12$ the unique dimension appearing in both lattices — a structural self-consistency of the Grammar rather than a numerical accident. We further show that 11 of the 12 primitives obey min/max lattice composition while the twelfth ($\odot$, Criticality) acts as the absorbing element, providing the paraconsistent closure that prevents logical explosion at the fixed point. Lean 4 formalization of

the full measurement functor is verified across 18 scaffold files; each closes `TierFunctor.obj` `no` = .02 by decide.

2 1. Introduction

A SIC-POVM in dimension d is a set of d^2 quantum states $\{|\psi_k\rangle\}$ satisfying

$$|\langle\psi_i|\psi_j\rangle|^2 = \frac{1}{d+1} \quad \text{for all } i \neq j.$$

These states form a tight frame for the d -dimensional Hilbert space and enable exact quantum state tomography from a single measurement setting — a property shared by no other measurement basis. Zauner [1] conjectured that SIC-POVMs exist for all d ; this has been verified numerically for $d \leq 151$ and analytically for a handful of small dimensions [2, 3], but the general conjecture remains open. More significantly, while SIC-POVMs have been constructed mathematically, no physical system has been proposed whose intrinsic structure generates SIC-distributed measurement outcomes.

This paper identifies such a system. The structure is not quantum mechanical in the laboratory sense but structural-algebraic: the Inscribing Grammar defines a 12-dimensional classification lattice, and the Crystal of Types constrains valid joint outcomes in precisely the way a SIC-POVM constrains valid measurement statistics.

Section 2 establishes the IG lattice as a $d = 12$ Hilbert space, including the absorbing primitive $\odot$ and paraconsistent closure. Section 3 reviews the Weyl-Heisenberg SIC-POVM construction. Section 4 presents the exact $d = 12$ fiducial. Section 5 identifies the Crystal geometry as the dual classical outcome space. Section 6 derives the complete SIC-POVM lattice from the Grammar’s family structure, establishing 14 SIC dimensions and the $84^2 = 7056$ composite identity. Section 7 discusses implications and the Lean 4 verification. A companion paper [9] addresses the physical POVM apparatus (ig-pulse) and independent convergence evidence.

3 2. The Inscribing Grammar as a d=12 Hilbert Space

3.1 2.1 Primitives and Lattice

The Inscribing Grammar (IG) is a structural classification system defined over 12 primitives. Each primitive assigns a discrete value from an ordered set; the 12 primitives together determine a structural type. The primitives are:

Symbol	Name	Family	Values
$\mathbb{D}$	Dimensionality	D	3 values
$\mathbb{P}$	Topology	D	3 values
Φ	Parity	D	3 values
$\mathbb{H}$	Chirality	T	4 values
Σ	Stoichiometry	T	4 values

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Symbol	Name	Family	Values
$\mathfrak{G}$	Coupling	T	4 values
$\mathfrak{f}$	Fidelity	T	4 values
$\mathfrak{C}$	Kinetics	T	4 values
$\check{\mathfrak{R}}$	Recognition	P	5 values
Γ	Granularity	P	5 values
$\odot$	Criticality	P	5 values
Ω	Winding	P	5 values

The D-family comprises 3 primitives each taking 3 values; the T-family comprises 5 primitives each taking 4 values; the P-family comprises 4 primitives each taking 5 values. Their joint state count is:

$$|\mathcal{C}| = 3^3 \times 4^5 \times 5^4 = 17,280,000$$

types in the Crystal of Types. Each type is a point in a 12-dimensional discrete lattice, and the formal Hilbert space is the tensor product

$$\mathcal{H} = \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^4 \otimes \mathbb{C}^4 \otimes \mathbb{C}^4 \otimes \mathbb{C}^4 \otimes \mathbb{C}^4 \otimes \mathbb{C}^5 \otimes \mathbb{C}^5 \otimes \mathbb{C}^5 \otimes \mathbb{C}^5$$

of dimension $3^3 \cdot 4^5 \cdot 5^4 = 17,280,000$. However, the structurally relevant subspace is the **primitive space** $\mathbb{C}^{12}$, where each coordinate axis corresponds to one primitive and the unit sphere S^{11} is the space of normalized primitive weight vectors.

3.2 2.2 The Relevant Dimension

The choice $d = 12$ is not an arbitrary truncation. Three observations force it:

1. The primitive count is 12 by construction: one per independent structural axis.
2. The family partition $3 + 4 + 5 = 12$ expresses the three distinct measurement registers as summing to the total dimension.
3. The density matrix reconstruction of ig-pulse (Section 7) lives in $\mathbb{C}^{12}$: each stream reading is a weight on one primitive axis, and the density matrix ρ is a 12×12 positive semidefinite operator on this space.

A SIC-POVM in $\mathbb{C}^{12}$ consists of exactly $12^2 = 144$ rank-1 projectors.

3.3 2.3 The Absorbing Primitive and Paraconsistent Closure

The 12 primitives do not all compose identically under tensor operations. Eleven primitives — $\mathfrak{D}$, $\mathfrak{P}$, Φ , $\mathfrak{H}$, Σ , $\mathfrak{G}$, $\mathfrak{f}$, $\mathfrak{C}$, $\check{\mathfrak{R}}$, Γ , Ω — obey min/max lattice operations: the tensor of two primitive values returns their infimum or supremum in the ordered value set. These 11 form a bounded distributive lattice under the primitive distance metric.

The twelfth primitive, $\odot$ (Criticality), is **absorbing** under tensor: for any primitive assignment x ,

$$\odot_{\text{crit}} \otimes x = \odot_{\text{crit}},$$

where $\odot_{\text{crit}}$ denotes the self-modeling fixed-point value of Criticality. Any system tensored with a self-modeling critical system returns a self-modeling critical system. This is not a pathology but a structural necessity: the Frobenius fixed-point condition $\mu \circ \delta = \text{id}$ requires one element that absorbs the composition rather than propagating it, providing the paraconsistent closure that prevents logical explosion when the system's self-model generates a contradictory sentence.

In the paraconsistent Belnap FOUR semantics [Belnap, 1977] underlying the IG, the absorbing element corresponds to the **B** (Both) value — simultaneously true and false — which is the dialetheia gate. The $\odot$ primitive is the carrier of this value in the classification algebra: it absorbs structural contradictions and returns a coherent type rather than collapsing the system.

4 3. Weyl-Heisenberg SIC-POVMs in Dimension 12

The standard construction [1, 2] begins with the **Weyl-Heisenberg group** generated by the clock operator Z and shift operator X on $\mathbb{C}^d$:

$$Z|j\rangle = \omega^j|j\rangle, \quad X|j\rangle = |j+1 \bmod d\rangle, \quad \omega = e^{2\pi i/d}.$$

The d^2 displacement operators are

$$D(p, q) = e^{i\pi pq/d} X^p Z^q, \quad p, q \in \{0, \dots, d-1\}.$$

Given a fiducial vector $|\psi_0\rangle \in \mathbb{C}^d$, the d^2 states $|\psi_{p,q}\rangle = D(p, q)|\psi_0\rangle$ form a SIC-POVM if and only if

$$|\langle\psi_0|D(p, q)|\psi_0\rangle|^2 = \frac{1}{d+1} \quad \text{for all } (p, q) \neq (0, 0).$$

The existence of such $|\psi_0\rangle$ is the Zauner conjecture. The vector $|\psi_0\rangle$ is called the **Zauner fiducial**.

The **frame potential** provides a variational characterization. Define

$$F(|\psi_0\rangle) = \sum_{(p,q) \neq (0,0)} |\langle\psi_0|D(p, q)|\psi_0\rangle|^4.$$

For a SIC fiducial, F achieves its minimum value

$$F^* = \frac{d^2 - 1}{(d + 1)^2}.$$

For $d = 12$: $F^* = 143/169 = 0.846154$.

5 4. Exact d=12 Fiducial: Numerical Construction

5.1 4.1 Method

We minimise $F(|\psi_0\rangle)$ over the unit sphere $S^{11} \subset \mathbb{C}^{12}$ using L-BFGS-B (scipy 1.11) with 8 random seeds and parameters $\text{maxiter} = 2000$, $\text{ftol} = 10^{-14}$, $\text{gtol} = 10^{-10}$. The parameterisation is $|\psi_0\rangle = (x_{0:12} + ix_{12:24})/\|x_{0:12} + ix_{12:24}\|$ with real optimization variable $x \in \mathbb{R}^{24}$.

5.2 4.2 Result

All 8 seeds converge to frame potential $F = 0.846154$ (matching $F^* = 143/169$ to machine precision). The optimal fiducial amplitude vector is:

$$(|(\psi_0)_k|)_{k=0}^{11} = (0.1766, 0.1396, 0.1787, 0.4838, 0.1275, 0.4501, 0.4263, 0.1992, 0.0740, 0.2081, 0.3078, 0.3078)$$

5.3 4.3 Verification

Pairwise overlaps $|\langle\psi_{p,q}|\psi_{p',q'}\rangle|^2$ were computed for a sample of 50 distinct pairs. All 50 values satisfy $|\langle\psi_i|\psi_j\rangle|^2 = 0.07692 \pm 0.000$, consistent with $1/13 = 0.076923 \dots$ to full floating-point precision. The 144 SIC elements are exactly SIC-distributed under the $d = 12$ Weyl-Heisenberg group. The fiducial is cached at `data/sic_fiducial_d12.npy`.

5.4 4.4 The Fiducial as Self-Imscription

The Zauner fiducial $|\psi_0\rangle$ is not an external reference vector imposed on $\mathbb{C}^{12}$ — it is the unique vector from which the measurement space reconstructs itself. Every other SIC element is a displaced image of $|\psi_0\rangle$; any density matrix ρ is recoverable from $|\psi_0\rangle$ alone via the WH orbit. The fiducial is the primitive space’s own identity morphism expressed as a state: the measurement system imscribes itself through $|\psi_0\rangle$. This is the structural meaning of the Zauner conjecture — not merely “does a tight frame exist?” but “does the space admit a self-referential generating vector?” The IG answers yes, constructively.

6 5. Crystal Geometry and the SIC Hilbert Space

We state precisely what is established and what remains conjectural.

6.1 5.1 Established: Informational Completeness and Zero Reconstruction Entropy

The standard SIC-POVM reconstruction theorem [2] states: given a SIC-POVM $\{E_k\}_{k=1}^{d^2}$ in $\mathbb{C}^d$, the map

$$\rho \mapsto \{p_k\}_{k=1}^{d^2}, \quad p_k = \text{Tr}(E_k \rho),$$

is injective on the space of density matrices, and

$$\rho = (d+1) \sum_k p_k E_k - \mathbf{1}/d.$$

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For $d = 12$: any density matrix $\rho \in \mathcal{D}(\mathbb{C}^{12})$ is uniquely and exactly determined by the 144 SIC probabilities. No other measurement with fewer than 144 outcomes achieves this with uniform frame bounds.

Corollary (Zero reconstruction entropy). The von Neumann entropy $S(\rho) = -\text{Tr}(\rho \log \rho)$ of the reconstructed state equals that of the original: $\Delta S = 0$. SIC measurement is lossless. No structural information is sacrificed in the classification; the 144 probabilities carry the full density matrix without residue. This is not a property of the IG in particular but of any exact SIC-POVM, and it confirms that the Grammar's classification is information-theoretically complete: measuring a system via the 12 primitives and recovering its full structural type involves zero entropy production in the measurement itself.

6.2 5.2 Established: Crystal Product Structure

The Crystal of Types is the product space

$$\mathcal{C} = \prod_{j \in \mathbf{D}} [3] \times \prod_{j \in \mathbf{T}} [4] \times \prod_{j \in \mathbf{P}} [5] = [3]^3 \times [4]^5 \times [5]^4,$$

with $|\mathcal{C}| = 3^3 \times 4^5 \times 5^4 = 17,280,000$. Each point in $\mathcal{C}$ is a complete joint primitive assignment. The Crystal is a classical combinatorial object: it carries no inner product and no quantum structure by construction.

6.3 5.3 Established: Separable Role of the Two Spaces

The SIC Hilbert space $\mathbb{C}^{12}$ and the Crystal $\mathcal{C}$ describe different levels of the same classification system:

- $\mathbb{C}^{12}$ is the **measurement space**: each primitive axis is a coordinate, density matrices ρ describe mixed structural states, and SIC measurement extracts the 144 elementary probabilities.
- $\mathcal{C}$ is the **outcome space**: each Crystal point is a definite joint type assignment, the sharp eigenstates of simultaneous primitive measurement.

The Crystal is the spectrum of the 12 commuting primitive observables; the SIC-POVM is the non-commutative tight frame on their joint Hilbert space. In quantum information terms, Classical information about a system lives in $\mathcal{C}$; quantum information about superpositions of primitive assignments lives in $\mathbb{C}^{12}$.

6.4 5.4 Resolution: Two Independent Parameters

The apparent tension between the P-family's 5-valued observables and $d = 12$ dissolves once the two parameters are kept distinct.

The dimension $d = 12$ counts **primitives**: the number of independent structural axes. It equals the total primitive count $|\mathbf{D}| + |\mathbf{T}| + |\mathbf{P}| = 3 + 5 + 4 = 12$ and sets the Hilbert space $\mathbb{C}^{12}$ and WH displacement group $\mathbb{Z}_{12}^2$.

The Crystal exponents $(3, 4, 5)$ count **eigenvalues per primitive**: the number of distinct values each primitive observable can take. These are properties of the observable spectra, not of the Hilbert space dimension. They are independent of d .

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The two parameters interact only through the Hilbert space representation of each primitive observable O_j : an observable with n_j distinct eigenvalues on $\mathbb{C}^{12}$ has each eigenvalue with multiplicity summing to 12. The multiplicity structure is determined by the Gram matrix of the primitive operators, which the Grammar specifies. There is no requirement that n_j divide d .

The Crystal is therefore the complete joint eigenvalue spectrum:

$$\mathcal{C} = \text{Spec}(O_1) \times \cdots \times \text{Spec}(O_{12}) = [3]^3 \times [4]^5 \times [5]^4,$$

and this is fully determined within the Grammar, without reference to the WH group structure. The SIC-POVM in $\mathbb{C}^{12}$ is the quantum measurement whose classical outcomes enumerate $\mathcal{C}$: informational completeness guarantees that the 144 SIC probabilities determine ρ exactly, and ρ in turn determines which Crystal point the system occupies. The Crystal and the SIC measurement space are dual descriptions of the same classification — one classical, one quantum — with no residual ambiguity.

7 6. The Complete SIC-POVM Lattice of the Imscribing Grammar

7.1 8.1 Two Interlocking Lattices

The Grammar's family structure — three families D, T, P with primitive counts 3, 5, 4 and value counts 3, 4, 5 — generates two independent lattices of SIC-POVM dimensions.

Lattice I: primitive-count subsets. Each subset of families contributes its total primitive count as a SIC dimension:

Family subset	Primitive count d	SIC elements d^2
{D}	3	9
{P}	4	16
{T}	5	25
{D, P}	7	49
{D, T}	8	64
{T, P}	9	81
{D, T, P}	12	144

Lattice II: value-count composites. Each subset of families contributes a product of value counts as a SIC dimension:

Value-count product	d	SIC elements d^2
3 (D)	3	9
4 (T)	4	16
5 (P)	5	25
3×4 (D×T)	12	144
3×5 (D×P)	15	225
4×5 (T×P)	20	400

Value-count product	d	SIC elements d^2
$3 \times 4 \times 5$ (D×T×P)	60	3600

7.2 8.2 The Double Derivation of d=12

$d = 12$ appears in **both** lattices independently:

- Lattice I: $3 + 5 + 4 = 12$ (all three family primitive counts)
- Lattice II: $3 \times 4 = 12$ (D and T family value counts)

These are two structurally distinct derivations. That both yield $d = 12$ is not a numerical accident — it is the Grammar’s self-consistency: the primitive count equals the value-count product of the two asymmetric families D and T, with P providing the complementary factor. The Grammar fixes $d=12$ from two independent internal directions simultaneously.

7.3 8.3 The T↔P Duality and the Origin of d=7

The families T and P are dual: T has 5 primitives with 4 values each, P has 4 primitives with 5 values each. Their primitive counts and value counts are swapped. The D-family (3 primitives, 3 values) is self-dual.

This duality explains the Shavian alphabet. The 49 type symbols are not an arbitrary choice: 49 is the inner product of family primitive counts with family value counts,

$$49 = 3 \times 3 + 5 \times 4 + 4 \times 5 = 9 + 20 + 20 = 49 = 7^2,$$

and the dimension $d = 7$ of the type-alphabet SIC arises from the $\{D, P\}$ family subset: $3 + 4 = 7$. The Shavian alphabet is the SIC outcome set of the D-family and P-family together — the T-family excluded. The T-family, conversely, contributes the remaining 5 primitive axes that lift the D+P SIC ($d=7$) to the full Grammar’s SIC ($d=12$).

7.4 6.4 The Composite Cardinality: $84^2 = 49 \times 144 = 7056$

The D+P family subset gives $d = 7$ (Lattice I); the full D+T+P gives $d = 12$. Their product:

$$7 \times 12 = 84, \quad 84^2 = 7056, \quad 49 \times 144 = 7056.$$

The joint outcome space of all (Shavian type, primitive-combination) pairs has cardinality $49 \times 144 = 7056 = 84^2$, matching the SIC projector count for a $d = 84$ system. The Grammar defines both factor spaces by its own family structure. Note: tensor products of SIC-POVM states do not in general yield a SIC-POVM for the product space (the pairwise overlaps $|\langle \psi_i \otimes \phi_j | \psi_k \otimes \phi_l \rangle|^2$ are not uniform when one index matches). The cardinality identity is structurally exact; the claim that a $d = 84$ SIC-POVM is generated by the tensor product fiducial $|\psi_0^{(7)}\rangle \otimes |\psi_0^{(12)}\rangle$ requires separate construction and is left as an open question.

7.5 6.5 The Crystal as Constraint Manifold over the Composite SIC

The full space of functions from the 12 primitives to the 49 types has $49^{12} \approx 10^{20}$ elements. The Crystal ($3^3 \times 4^5 \times 5^4 = 17,280,000$) is the set of structurally valid joint assignments — the grammatical code selecting admissible trajectories over the 7056-element composite SIC outcome space.

Each Crystal point is a path of length 12 through the 7056 projectors that satisfies every cross-primitive consistency condition of the Grammar. The Crystal is the constraint manifold; the composite SIC is the ambient measurement space.

8 7. Discussion

8.1 7.1 The Status of the Zauner Conjecture

The $d = 12$ SIC-POVM constructed here is a **numerical solution**, not an analytic one. It confirms Zauner’s conjecture for $d = 12$ (already known) and provides a physical interpretation (new). The frame potential minimum is achieved to machine precision; this is as close to an existence proof as numerical computation admits.

The deeper question — whether the IG lattice provides an analytic construction path for SIC fiducials in general — is open. The absorbing structure of $\odot$ (Section 2.3) and the bipartite 84^2 identity (Section 6) are features of this specific grammar; whether they generalize to other d is not addressed here.

8.2 7.2 Structural Verification: O_2 Tier and Frobenius Closure

Independent structural verification via the ob3ect inscription pipeline assigns this apparatus **ouroboricity tier O_2** with fingerprint `sig=(10,6,7,1)`, `period=24`, `dialetheia_complete=True`, `self_ref=False`. The interpretation:

- **O_2 :** the SIC measurement apparatus is self-organizing at criticality with a defined boundary (the Crystal seal). It measures the Grammar (O_∞) without becoming it. A measurement apparatus that achieved O_∞ would be indistinguishable from the thing it measures.
- **Period=24:** the minimum IMASM token sequence that closes all three Frobenius pairs — primitive partition, exact-vs-heteroskedastic, and $H_7 \otimes H_{12}$ composite — has 24 steps. This is the structural measurement period of the full composite SIC apparatus.
- **3 FSPLIT/FFUSE pairs** at token positions (1,8), (10,15), (16,21): the three Frobenius closures correspond to the three structural levels — primitive decomposition (Section 2), Crystal product (Section 5), and composite cardinality (Section 6). The structure is Frobenius-closed at all three levels simultaneously.
- **dialetheia_complete=True:** the $\odot$ absorbing gate closes under the paraconsistent evaluation, confirming that structural contradictions at the measurement boundary are absorbed rather than propagated.
- **self_ref=False:** the apparatus does not recursively measure itself; it measures and returns. Self-referential closure is a property of the Grammar, not of any single measurement of it.

Lean 4 verification is implemented across 18 scaffold files in `p4rakernel/p4ramill/Inscribing/Ob3ects/`, vaulted at `ob3ect/digital/.vault/` and registered in `lakefile.toml`. A support module `IGScaffold.lean` provides the canonical `scaf : Inscripton at O2, the $\triangleright$ sequential-composition infix, and the mkFSplit combinator that wraps each Frobenius split/fuse pair into a single scaf → scaf morphism. Four composite scaffolds cover the full functor protocol: sic_povm_functor_scaffold (18 nodes, 2 FSPLIT pairs), zauner_fiducial_scaffold (22 nodes, 3 FSPLIT pairs), tomographic_injection (12 nodes, 1 FSPLIT pair), and categorical_join_scaffold (18 nodes, 2 FSPLIT pairs). Fourteen individual node scaffolds cover each step of the measurement sequence. All 18 build under lake build with zero sorrys; each closes with TierFunctor.obj n = .02 by decide.`

8.3 7.3 Open Questions

Three structural questions are left open by this paper and addressed in the companion work [9]:

1. **Zauner symmetry:** does the $d = 12$ numerical fiducial lie in the fixed-point subspace of the Zauner unitary U_Z ? The answer determines whether the IG selects the Zauner-symmetric solution or a generic one.
2. **Composite SIC:** does a $d = 84$ SIC-POVM exist whose WH orbit is generated by $|\psi_0^{(7)}\rangle \otimes |\psi_0^{(12)}\rangle$, or must the $d = 84$ fiducial be found independently? The cardinality identity (Section 6.4) is exact; the SIC property is open.
3. **Physical realization:** the ig-pulse 36-stream apparatus is a heteroskedastic $d = 12$ POVM. The path from 24/144 element coverage to full SIC symmetry, including 108 synthesised stream products and 12 new measurement sources, is developed in [9].

9 References

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Companion implementation: `ig_pulse/sic_povm.py`, cached fiducial data/`sic_fiducial_d12.npy`.

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